



Hệ thống kiểm tra trùng lặp nội dung

KẾT QUẢ KIỂM TRA TRÙNG TÀI LIỆU

THÔNG TIN TÀI LIỆU

Tên tác giả	SuperAdmin
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ĐOẠN VĂN TRÙNG LẶP

```
#include "glut.h"
#include <math.h>
void DrawCoordinate()//vẽ hệ trục tọa độ xy
//Vẽ trục x - màu xanh lá
glColor3f(0.0, 1.0, 0.0);
glBegin(GL_LINES);
glVertex3f(-1.0, 0.0, 0.0);
glVertex3f(1.0, 0.0, 0.0);
//Vẽ trục y - màu đỏ
glColor3f(1.0, 0.0, 0.0);
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glBegin(GL_LINES);
glVertex3f(0.0, -1.0, 0.0);
glVertex3f(0.0, 1.0, 0.0);
// --- Inserted DrawBookshelf function here ---
void DrawBookshelf()
// Draw bookshelf background (brown)
glColor3f(0.6f, 0.3f, 0.1f);
glBegin(GL_POLYGON);
glVertex2f(-0.7f, -0.7f);
glVertex2f(0.7f, -0.7f);
glVertex2f(0.7f, 0.7f);
glVertex2f(-0.7f, 0.7f);
// Draw bookshelf border (darker brown)
glColor3f(0.4f, 0.2f, 0.05f);
glLineWidth(8.0f);
glBegin(GL_LINE_LOOP);
glVertex2f(-0.7f, -0.7f);
glVertex2f(0.7f, -0.7f);
glVertex2f(0.7f, 0.7f);
glVertex2f(-0.7f, 0.7f);
// Draw three horizontal shelf lines (creates 4 shelves)
glColor3f(0.4f, 0.2f, 0.05f);
glLineWidth(4.0f);
glBegin(GL_LINES);
glVertex2f(-0.7f, 0.25f); glVertex2f(0.7f, 0.25f); // 1st line (top shelf)
glVertex2f(-0.7f, -0.15f); glVertex2f(0.7f, -0.15f); // 2nd line (middle shelf)
glVertex2f(-0.7f, -0.55f); glVertex2f(0.7f, -0.55f); // 3rd line (bottom shelf)
// Top shelf: blue circle (left)
glColor3f(0.4f, 0.8f, 1.0f);
float cx = -0.45f, cy = 0.5f, r = 0.07f;
glBegin(GL_POLYGON);
for (int i = 0; i < 40; ++i) {
float theta = 2.0f * 3.1415926f * float(i) / 40.0f;
glVertex2f(cx + r * cosf(theta), cy + r * sinf(theta));
// Top shelf: pink triangle (center)
glColor3f(1.0f, 0.6f, 0.9f);
glBegin(GL_TRIANGLES);

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glVertex2f(0.0f, 0.55f);
glVertex2f(0.12f, 0.38f);
glVertex2f(-0.12f, 0.38f);
// Bottom shelf: three vertical green books (left)
glColor3f(0.3f, 1.0f, 0.5f);
float bx = -0.55f, by = -0.65f, bw = 0.05f, bh = 0.25f;
for (int i = 0; i < 3; ++i) {
glBegin(GL_POLYGON);
glVertex2f(bx + i * 0.09f, by);
glVertex2f(bx + i * 0.09f + bw, by);
glVertex2f(bx + i * 0.09f + bw, by + bh);
glVertex2f(bx + i * 0.09f, by + bh);
// Bottom shelf: slanted green book (left)
glPushMatrix();
glTranslatef(-0.35f, -0.55f, 0.0f);
glRotatef(-25, 0, 0, 1);
glBegin(GL_POLYGON);
glVertex2f(-0.025f, 0.0f);
glVertex2f(0.025f, 0.0f);
glVertex2f(0.025f, 0.18f);
glVertex2f(-0.025f, 0.18f);
glPopMatrix();
// Third shelf: slanted green book (right)
glPushMatrix();
glTranslatef(0.45f, 0.15f, 0.0f);
glRotatef(25, 0, 0, 1);
glBegin(GL_POLYGON);
glVertex2f(-0.025f, 0.0f);
glVertex2f(0.025f, 0.0f);
glVertex2f(0.025f, 0.28f);
glVertex2f(-0.025f, 0.28f);
glPopMatrix();
void RenderScene()
glClear(GL_COLOR_BUFFER_BIT);
glLoadIdentity();
DrawCoordinate(); // Draw coordinate axes
// Draw the bookshelf

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DrawBookshelf();
// Save the current coordinate system
glPushMatrix();
// Translate the coordinate system
glTranslatef(0.2, 0.0, 0.0);
// Rotate the coordinate system (e.g., -30 degrees for clockwise rotation)
glRotatef(-30, 0.0, 0.0, 1.0);
// Scale the coordinate system
glScalef(0.5, 0.5, 0.5);
// (Polygon drawing code removed)
// Restore the previous coordinate system
glPopMatrix();
glutSwapBuffers();
void init(void)
glClearColor(1.0, 1.0, 1.0, 1.0); /* nền đen */
glOrtho(-1.0, 1.0, -1.0, 1.0, -1.0, 1.0);
void main()
glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
glutInitWindowSize(640, 480);
glutInitWindowPosition(100, 100);
glutCreateWindow("Translate");
glutDisplayFunc(RenderScene);
glutMainLoop();
```